

Yinsheng Yao

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EDUCATION

Tongji University

Bachelor of Engineering in Computer Science and Technology

GPA: 89.38/100

Selected Honors: Second Prize, Tongji University Scholarship (2022-2023 Academic Year)

Shanghai, China

Sep. 2022 – Jun. 2026

RESEARCH EXPERIENCE

Hierarchical Benchmark for Automated Financial Research Reporting

Oct. 2025 – Present

Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Designed and constructed a benchmark for deep analysis of financial research reports, targeting causal reasoning, factual grounding, risk identification, and comparative analysis in long-form financial texts.
- Led the development of the financial deep analysis module, defining fine-grained evaluation dimensions and building a structured QA dataset with rule-based and numerical reasoning challenges.
- Contributed to [a paper](#) based on this benchmark.

Multi-Agent Verification-Elaboration Network with In-Step Epistemic Auditing

Aug. 2025 – Present

Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Designed and implemented a multi-agent reasoning architecture integrating specialized roles (Planner, Researcher, Skeptic, Judge, and Stylist) to support multi-stage planning, internal debate, external verification, and consensus synthesis for complex reasoning tasks.
- Developed a reasoning-hop-based routing mechanism, enabling dynamic path selection between fast and full reasoning pipelines based on task complexity; Implemented asynchronous web verification and structured deliberation loops to improve factual consistency and causal reasoning.
- Conducted automated evaluations on OBQA, TruthfulQA, HALUEVAL and StrategyQA datasets, achieving great performance compared with the baseline.
- Contributed to writing [a research paper](#) based on this framework as the first author.

Fine-Grained Diagnostic Feedback for LLM Code Refinement

May 2025 – Present

Group Project, Supervisor: Prof. Tianyi Zhang, Purdue University

- Proposed FLARE, an iterative LLM code refinement framework combining line-level diagnostic feedback with execution-based candidate verification.
- Evaluated on LiveCodeBench and BigCodeBench across five base LLMs; FLARE outperformed Self-Debugging, Self-Refine, and NL-Debugging in most settings.
- Achieved 1.72%–7.42% absolute Pass@1 gains over the strongest baseline without candidate search, plus 8.50% average improvement with top-10 search.
- Contributed to writing [a research paper](#) based on this framework as the first author.

Chinese Character Style Recognition System

Jan. 2023 – Mar. 2026

Group Project, Supervisor: Prof. Chen Ye, Tongji University

- Enhanced discrimination through ArcFace loss function, improving the separation of visually similar calligraphic styles
- Constructed a benchmark comparing CNN and ViT architectures under simple vs. strong augmentation, providing reproducible baselines for style recognition.
- Built HCSU, the first fine-grained calligraphy dataset with 39,307 images from 49 calligraphers across 10 dynasties, decoupling ink manuscripts (Tie) from stone rubbings (Bei) to resolve modal mixture; provided hierarchical aesthetic descriptions for style discrimination and aesthetic reasoning evaluation.
- Authored [a paper](#) (ECCV 2026, first author) revealing a knowledgeable but unperceptive paradox in state-of-the-art LVLMs on cultural heritage visual tasks.

Other Project Deliverables and Achievements:

- Submitted a deployed system as a project and received the Excellent Project Award in the University Innovation and Entrepreneurship Training Program.
- Presented project at the “Between the Lines: The Beauty of Chinese Characters and the Path of Intelligent Evolution” exhibition at Tongji University Museum.
- Relevant work was accepted by the 1st CCF AI for Humanities Conference (Beijing, China, 2025); the dataset was selected by the conference as a recommended dataset case (Top 40).

PROFESSIONAL SKILL

Programming: Python, C, C++ (Proficient in syntax and object-oriented design), MATLAB, Verilog, Assembly

Frameworks & Tools: PyTorch, timm, Docker, GitHub, Hugging Face

Database & Back-end: MySQL (with basic optimization), OceanBase, MiniOB

Front-end Development: HTML5, CSS, JavaScript

AI Application: Deep Learning, Computer Vision, Transformer Models, Large Language Models, Prompt Engineering